

ADARSH GUPTA

Bangalore, Karnataka | +91 6306297757 | itsmeag04@gmail.com | [Linkedin](#) | [Github](#)

EDUCATION

Indian Institute of Technology, Dharwad 2022 – 2026

B.Tech. in Mathematics and Computing | CPI: 8.88 / 10.0

Relevant Coursework: Data Structures & Algorithms, Distributed Systems, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems, Linear Algebra, Optimization Theory

EXPERIENCE

Arctan Technologies

Jan 2026 – April 2026

Software / ML Engineering Intern

Bangalore, India

- Reduced agent onboarding time by 60% by designing and shipping an end-to-end Voice Roleplay Bot with automated scoring, naturalized speech pattern processing for numeric data, and real-time feedback delivered in an agile 3-week sprint.
- Engineered scalable synthetic-data pipelines using TTS and Dynamic Time Warping (DTW) for frame-level alignment, generating 100+ hours of parallel training audio across distributed computing environments.
- Built an accent-transfer system from scratch, implementing latent-space mapping of S3 tokens in CosyVoice and fine-tuning SeedVC diffusion layers for Indian speech, optimizing and separating the pipelines for real-time versus offline inference.

Dvara E-Registry

Jun 2025 – Dec 2025

Software / ML Engineering Intern

Dharwad, India

- Reduced commodity price-forecasting error to 10% MAPE (14-day horizon) and 14% MAPE (4-week), outperforming baselines by 25% by engineering hybrid LightGBM + ARIMA/SARIMA models incorporating 365-day daily seasonality, PACF-based lag features, and nonlinear optimization.
- Built multi-tiered, district-level price-forecasting pipelines on large-scale datasets, aggregating multi-market daily feeds and modeling volatility trends using LSTM and LightGBM handling ambiguous, undefined requirements end-to-end.
- Articulated complex forecasting insights and technical trade-offs to non-technical stakeholders, enabling data-driven procurement decisions across 15 districts.

PROJECTS

Live Polling Platform

Dec 2025 – Jan 2026

React · Node.js · Socket.io · TypeScript · Docker · Nginx

- Built a fault-tolerant, distributed live-polling system supporting 500 concurrent users, achieving sub-100ms broadcast latency via bidirectional WebSocket communication with Socket.io on an Express backend.
- Designed scalable, multi-tiered deployment using Docker and Nginx, cutting estimated hosting cost by 40% versus a single-server architecture while maintaining high availability.
- Implemented role-based access and relational database query schemas for session management, enabling instructors to moderate 20 concurrent sessions with real-time participation analytics.

Cricket Fantasy Analytics Pipeline

Nov 2024 – Jan 2025

Python · Data Engineering · XGBoost · LSTM · SHAP · LLM APIs

- Architected an end-to-end automated data ingestion pipeline integrating multi-source API feeds (Cricsheet, Cricbuzz) to process domain-specific features and train predictive models via nonlinear optimization.
- Built an inference engine integrating SHAP values and LLaMA-generated explanations, making complex backend outputs highly interpretable and actionable for non-technical end users.

Peer-to-Peer VoIP Platform

Mar 2024 – Apr 2024

Python · Sockets · UDP/TCP · PyAudio · Multithreading

- Architected a decentralized, distributed peer-to-peer VoIP system using object-oriented design, achieving low-latency audio streaming for up to 8 concurrent peers over UDP with thread-safe fault-tolerant error recovery.
- Engineered multithreaded tracker server handling peer registration enabling reliable file sharing and direct audio streaming.

TECHNICAL SKILLS

Languages: C/C++, Python, Java, SQL, JavaScript, TypeScript, HTML/CSS

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Design, Distributed Systems, Relational Databases

Systems & Architecture: Docker, Linux, Bash, Git, FastAPI, Flask, RESTful API, WebSocket, Nginx, Distributed Computing

AI/ML: PyTorch, TensorFlow, Scikit-learn, Transformers, OpenCV, LLM Fine-tuning, RAG, Streamlit

Certifications: [Machine Learning Specialization \(Stanford / DeepLearning.AI\)](#), [Introduction to Data Science \(Cisco\)](#), [SQL Advanced \(HackerRank\)](#), [InterIIT Tech Meet 13.0 \(IIT Bombay\)](#)

LEADERSHIP

- Led IIT Dharwad's team at Inter-IIT Tech Meet 13.0 (IIT Bombay), coordinating 8 engineers across a 6-week competition sprint, managed ambiguous deliverables and articulated technical solutions to judging panels.